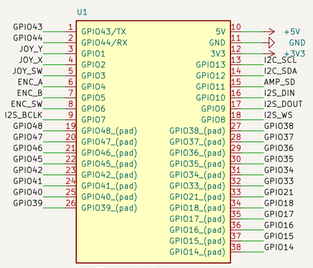
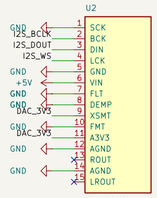
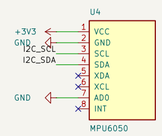
ESP32-S3 SuperMini (pogo pins reach the pads underneath)



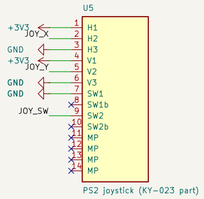
DAC (GY-PCM5102, line out; FLT DEMP FMT low, XSMT high)



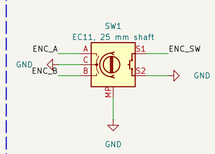
Tilt (GY-521 / MPU-6050, I2C 0x68)



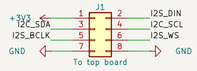
Joystick (the KY-023's PS2-style stick, no module board; press = record)



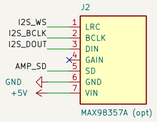
Encoder (EC11, shaft up through the top board)



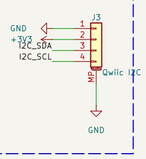
To the top board (OLED on I2C, mic on I2S)



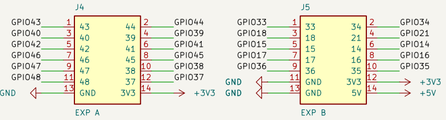
Amp header (optional MAX98357A)



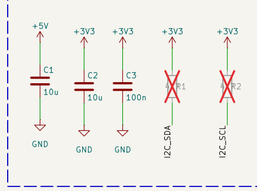
Qwiic



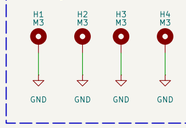
Breakouts (every spare GPIO)



Decoupling, I2C pull-ups



Mounting





USB-C on the SuperMini powers everything: its 5V pin feeds the DAC and the amp, its 3.3 V regulator feeds the IMU, joystick, top board (OLED, mic), Qwiic and the breakouts.

DAC control pins are tied here, no straps: FLT, DEMP, FMT to GND (normal filter, no de-emphasis, I2S), XSMT to A3V3, the module's own 3.3 V (unmuted). A floating FMT distorts at a touch. Matches the H1L-H4L bridges on the DAC board's back if they are set as usual (L L H L).

Joystick pots run from 3V3 (not 5V); the ESP32 ADC reads 0-3.3 V. The IMU's +X axis points at the nose (the USB / jack edge). AD0 to GND: address 0x68.

EXP A / EXP B carry every GPIO the design leaves free, including the SuperMini's underside pads (reached with pogo pins). 45 and 46 are strapping pins; 48 drives the module's RGB LED.

R1, R2 are not fitted: the OLED and GY-521 boards carry I2C pull-ups.

Mates with samplooler-stack-top through J1 (2x4, 11 mm)
ESP32-S3 SuperMini carrier: DAC, IMU, joystick, encoder, amp header, breakouts

Samplooler

Sheet: /
File: samplooler-stack-bottom.kicad_sch

Title: Samplooler stack, bottom board: modules

Size: A2	Date: 2026-10-15	**Rev: D**
KiCad E.D.A. 10.0.6		Id: 1/1